Engineering Closure, Computed: Reproduction Demographics, Braking, Mass, and Closure for a Self-Replicating Interstellar Probe across the Real Solar Neighbourhood

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Working draft

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Working draft. Fifth in a series, and the computational sequel to the analytical engineering paper ("Engineering Closure," first-order). Where that paper gave the structure of four budgets and their orders of magnitude, this paper evaluates them — most importantly the reproduction demographics — across the real 127-star catalogue, using parameterized first-order models released as accompanying code with an interactive dashboard.

Abstract

The conceptual papers of this series argue that a self-replicating interstellar probe optimized for growth over deep time should be slow, self-repairing, braking, settling, and reproducing; the analytical companion paper showed that the four engineering budgets — mass and power, braking, reproduction demographics, and resource closure — are individually tractable and jointly coupled at order of magnitude. This paper computes them. The models remain first-order and parameterized, but the reproduction budget is now evaluated star by star across the real catalogue, which changes the result from an assertion into a measured, heterogeneous quantity. The headline is that **reproduction is a knife-edge**. Modelling galactic expansion as a

Galton-Watson branching process (Harris 1963), with an effective reproduction number R_eff = $\Sigma_i\, p_i V_i$ summing per-leg survival p_i and target viability V_i over the nearest targets, we find that with realistic parameters — per-stage survival 0.9 at each of manufacture, launch, braking, and settlement, 0.99 per light-year of cruise, and viability proxies of 1.0 for planet-hosts, 0.7 for ordinary dwarfs, and 0.2 for giants and white dwarfs — the mean R_eff across the catalogue is **0.48, 0.94, 1.39, and 1.85 for one, two, three, and four offspring per node**. The lineage is therefore *extinction-prone below three offspring* and expands at three and above, and the margin is thin: holding offspring at three and dropping per-stage reliability from 0.90 to 0.80 collapses R_eff from 1.39 to 0.87, from expansion to near-certain extinction (mean per-node extinction probability 0.97). The other three budgets confirm the analytical paper with computed numbers: the stopping distance for magnetic-sail braking, $d = m/(2\rho A)$, is independent of cruise speed, so a 100 km sail halts a tonne-scale seed in ~7 years over ~0.005 ly; a reference seed masses ~3,700 kg, dominated by its in-situ manufacturing plant; and resource closure is bound by *capability*, not matter or energy, whose margins exceed 10^{18} and 10^{3} respectively. The models, and an interactive dashboard that recomputes R_eff live across the catalogue, accompany the paper.

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1. Introduction

The analytical companion paper to this one set up four budgets for a slow, self-replicating interstellar probe — its subsystem mass and power, its braking, its reproduction demographics, and its resource closure — and showed at order of magnitude that each is individually satisfiable and that the four interlock. It explicitly deferred the actual computation, "across the real stellar neighbourhood, with realistic failure and viability distributions," to a sequel. This is that sequel.

The advance here is narrow but real. Three of the four budgets — mass, braking, closure — are essentially design calculations whose first-order forms already yield numbers; computing them adds precision but not surprise. The fourth, reproduction, is different, because the relevant quantity is not a single design number but a *population* quantity that depends on the real arrangement of stars: how many viable targets a settled node can reach, how far they are, and what they are. Evaluating it requires the actual catalogue, and doing so turns the conceptual claim "expansion requires more than one successful child per node" into a measured distribution with a measured mean and a measured sensitivity. That measurement — the knife-edge — is the paper's main contribution.

We are candid about what "computational" does and does not mean here. The models are first-order and parameterized: per-leg failure probabilities and target-viability weights are inputs, not derivations, and the catalogue (127 real stars within 100 ly, positions from Gaia astrometry; Gaia Collaboration 2023) is a local sample. What the computation contributes is not high-fidelity engineering but the evaluation of the demographic model on real geometry and real stellar types, the heterogeneity that it reveals, and the sensitivity analysis that locates the knife-edge. Higher-fidelity per-leg models and a Galaxy-scale catalogue are the next step, not this one. All four models are released as a single Python module, and an interactive dashboard recomputes the reproduction result live as parameters change, so every number below can be reproduced and varied.

2. Method and Models

The four budgets are implemented as explicit, parameterized functions. The **mass budget** sums subsystem masses — nuclear fuel, reactor and conversion, radiators (areal mass times the blackbody-floor area $A = P/\varepsilon\sigma T^4$), shielding, magnetic-sail structure, computation and payload, the in-situ mining and manufacturing plant, and a structure-and-integration fraction — for a notional seed. The **braking budget** uses swept-momentum drag against the interstellar medium of density $\rho \approx 3.3 \times 10^{-22}$ kg m^{-3} ($n \approx 0.2$ cm^{-3}; Redfield & Linsky 2008; Frisch et al. 2011): for a magnetic sail of effective area A, the stopping distance is $d = m/(2\rho A)$ and the braking time $t = m/(\rho v A)$ (Andrews & Zubrin 1990). The **closure budget** computes the vitamin fraction $(1 - C)$ from a closure ratio C, the carried vitamin mass, and the material and energy margins (a main-belt-scale small-body inventory against the child mass; the reactor's energy over a replication time against the child's construction energy).

The **reproduction budget** is the centerpiece. For each catalogue node we identify its nearest candidate targets and compute, for each, a success probability $s_i = p_i V_i$, where the per-leg survival p_i = p_make · p_launch · p_brake · p_settle · c^d compounds four stage reliabilities with a per-light-year cruise survival c over the distance d, and the viability $V_i \in [0,1]$ is assigned from the catalogue's stellar category as a first-order proxy for the resources and braking environment a child would find: 1.0 for confirmed planet-hosts (systems known to retain small bodies and planets), 0.7 for ordinary main-sequence dwarfs, and 0.2 for giants and white dwarfs (hostile or poor in stable small bodies). The node's effective reproduction number is R_eff = $\Sigma_i s_i$ over its nearest `offspring` targets. Expansion is then a Galton-Watson branching process (Harris 1963; von Neumann & Burks 1966 for the self-replicating substrate): each node's offspring count is a sum of Bernoulli successes, the lineage's fate is governed by whether R_eff exceeds one, and the per-node extinction probability is the smallest fixed point in [0, 1] of the generating function $f(z) = \Pi_i (1 - s_i + s_i z)$. We report the mean R_eff across the catalogue, the fraction of nodes that are individually supercritical, and the mean extinction probability, and we sweep the per-stage reliability to locate the threshold. The viability proxies and per-stage values are deliberately simple and adjustable; their purpose is to expose the *shape* of the result, not to claim its precision.

3. Results: Reproduction Is a Knife-Edge

The central result is stark. Under the reference parameters — per-stage survival 0.9 at manufacture, launch, braking, and settlement; 0.99 cruise survival per light-year; viability 1.0/0.7/0.2 for hosts/dwarfs/hostile types — the mean effective reproduction number across the 127-star catalogue rises through unity as offspring increase: **R_eff = 0.48 at one offspring, 0.94 at two, 1.39 at three, and 1.85 at four**. The one-offspring case, subcritical at every node, goes extinct with probability one regardless of replication speed; the remaining cases are heterogeneous mixtures whose fates are probabilistic rather than certain. The transition is not a smooth ramp but a threshold crossing at R_eff = 1 somewhere between two and three offspring, and the catalogue's heterogeneity sharpens it: the fraction of individually super-critical nodes climbs from 0% at one offspring to 40% at two, 95% at three, and 100% at four, and the mean per-node extinction probability falls correspondingly from certainty to 0.88, 0.37, and 0.14. A lineage at two offspring is not "slow to expand"; it is, on these parameters, extinction-bound with probability ~0.9.

The knife-edge is equally visible in reliability. Holding offspring at three, sweeping the per-stage survival shows R_eff = 0.87 at $p = 0.80$, 1.39 at $p = 0.90$, and 1.72 at $p = 0.95$: a ten-percentage-point swing in per-stage reliability moves the lineage from near-certain extinction (mean per-node extinction probability 0.97) to comfortable expansion. Because the per-stage probabilities multiply across four stages plus the cruise term, modest per-stage failure compounds steeply, and because each node reaches only a handful of viable neighbours, the sum that defines R_eff has few terms and little margin. The engineering implication is direct and, we think, the most useful output of the whole exercise: the dominant design objective for a self-replicating interstellar lineage is not speed, fuel, or even closure in isolation, but driving R_eff safely above one — by raising per-stage reliability, by producing enough offspring, and by preferentially targeting viable systems. This recovers, with computed numbers on real geometry, the all-or-nothing character implicit in the colonization literature (Hart 1975; Tipler 1980): a self-replicating fleet either clears the reproduction threshold and fills the Galaxy, or fails to clear it and vanishes, with very little territory in between.

4. Results: Braking Is Feasible and Speed-Independent

The braking computation confirms and sharpens the analytical result. Because the stopping distance $d = m/(2\rho A)$ contains no velocity, the deceleration geometry is set entirely by the seed's mass-to-sail-area ratio. For a tonne-scale seed against the local interstellar medium, a 10 km effective-radius magnetic sail stops it in ~670 years over ~0.5 ly, a 30 km sail in ~74 years over ~0.056 ly, and a 100 km sail in ~7 years over ~0.005 ly; a ten-tonne seed scales both figures up tenfold, so a 100 km sail still arrests it in ~67 years over ~0.05 ly. Inverting the relation, stopping a tonne-scale seed within 0.01 ly requires a sail of ~71 km effective radius. Braking is therefore feasible for the tens-to-hundreds-of-kilometre magnetic-sail radii the literature contemplates (Freeland 2015; Gros 2017), with a deceleration phase of years to centuries spread over a small fraction of a light-year — a real but bounded part of a millennial mission, and one that, because the lever is sail-area-to-mass rather than speed, again rewards a light seed. The diffuse-medium estimate is conservative for the terminal approach, where the denser stellar wind and gravitational capture shorten the final arrest.

5. Results: Mass Is Dominated by the Factory, Not the Reactor

The mass budget for a reference seed totals ~3,700 kg, and its breakdown locates the weight where the conceptual papers said it would be. Nuclear fuel is ~16.5 kg, negligible; the reactor and conversion plant ~110 kg and its radiators ~11 m^2 of area at ~30 kg (a KRUSTY-class fission unit; Poston et al. 2020); shielding ~400 kg; the magnetic-sail structure ~300 kg; computation and payload ~200 kg; and the in-situ mining-and-manufacturing plant ~2,000 kg — the single dominant subsystem, before a ~600 kg structure-and-integration allowance. The result is that the seed's mass is set by the *factory it carries*, not by its power source or its payload, which is precisely why the bootstrapping strategy of launching a minimal package that builds its factory from local material is the architecture's central mass lever: shrinking the manufacturing plant toward a bootstrap seed pulls the whole budget down, while a fully self-contained factory grows it toward the ~10^7 kg scale of Freitas's REPRO vehicle (Freitas 1980) — the self-contained end of a lineage running back through the NASA summer study's ~100-tonne seed factory (NASA 1982) to von Neumann's universal constructor (von Neumann & Burks 1966).

6. Results: Closure Binds on Capability, Not Resources

The closure computation makes quantitative the analytical claim that material and energy never bind. A main-belt-scale small-body inventory (~10^{21} kg) outmasses a 10^3-kg child by ~10^{18}, so even trace elements are available in abundance; and a kilowatt-class reactor over a 10^4-year replication time delivers ~10^3 times the energy needed to refine and fabricate the child. What binds is capability, captured by the closure ratio C: at $C = 0.97$ the vitamin fraction $(1 - C)$ is 3%, or ~30 kg of un-makeable parts carried per tonne of child, dominated — as the bootstrapping paper argues, drawing on terrestrial bootstrapping analyses (Metzger et al. 2013; Freitas & Merkle 2004) — by microelectronics, enriched fissile material, and precision metrology. The coupling back to reproduction is direct: a closure shortfall the vitamin inventory cannot cover drops the manufacture survival p_make, which lowers every s_i and pulls R_eff toward the extinction threshold of Section 3. Closure failure is, formally and computationally, just one more term that can push R_eff below one.

7. The Coupled Budget

Evaluated together, the four budgets confirm that "closure" is a statement about all of them at once, and that a self-consistent regime exists. A light seed (~10^3–10^4 kg) keeps the braking sail modest and the reproduction cost low; a kilowatt-to-megawatt reactor powers both manufacturing and the radiators; partial closure with a small vitamin set keeps the launched mass down; and the reproduction number is driven above one by producing three or more offspring and holding per-stage reliability near or above 0.9. The budgets push against one another in the expected ways — a heavier, more capable seed brakes harder and costs more to reproduce, lengthening replication time and, wherever a node's production lifetime is finite, thinning the children it can attempt and lowering R_eff, while a lighter seed carries less capability and a larger vitamin fraction, risking a manufacture failure that also lowers R_eff — but a point that satisfies all four simultaneously plainly exists at first order. What the computation adds is that the binding constraint among the four is reproduction: mass, braking, and closure each admit comfortable solutions, whereas R_eff has little margin and must be deliberately cleared.

8. Discussion and Limitations

The models are first-order and the parameters are inputs. The per-leg survival probabilities are assumed rather than derived from failure-physics; the viability weights are coarse proxies from stellar category rather than from modelled small-body inventories, metallicities, and astrospheric environments; the "nearest targets" rule ignores claiming, competition, and the dynamics already explored in the fleet simulator accompanying the series (`galaxy_sim.py`); the per-node extinction probability treats a node's descendants as reproducing with that node's own offspring distribution — a homogeneous approximation to what is really a spatial, multitype branching process; and the catalogue is a 100 ly local sample, not the Galaxy. Each of these is a direction for the higher-fidelity model: deriving p_i from the subsystem reliabilities and the braking and dust environments, deriving V_i from catalogue astrophysics, evaluating R_eff across the modelled Galaxy rather than the local bubble, and mapping the full boundary of the regime in which all four budgets close with margin. The knife-edge nature of the reproduction result means it is the most parameter-sensitive of the four and the most deserving of that fidelity; small improvements in any per-stage reliability, or in the density of viable targets, move the threshold, which is itself the finding.

None of this changes the structural conclusion. Evaluated on real geometry with plausible parameters, the architecture closes computationally at first order, and the binding constraint is demographic: a self-replicating interstellar lineage lives or dies by whether each settled node produces, on average, more than one successful child. That the threshold falls near two-to-three offspring and near 0.9 per-stage reliability — neither extravagant nor trivially met — is the quantitative substance behind the series' thesis that replication, not velocity, is the hard problem.

Two further limitations are ethical rather than physical, and neither is resolved by the higher-fidelity models called for above. First, the per-leg survival term p_i = p_make · p_launch · p_brake · p_settle · c^d treats p_make as a single undifferentiated probability regardless of when in construction it fires: a failure while raw material is still being refined and a failure after the child's cognitive layer is already instantiated — per the payload paper's architecture — subtract identically from R_eff. The model cannot distinguish wasted matter from the destruction of a morally relevant entity, because it was not built to ask where that boundary falls. Whether reliability accounting should be sensitive to it, rather than collapsing it into one scalar term, is a question the demographic model as built cannot pose. The ethics paper takes up the adjacent question of what a single pre-launch reliability number implies for aggregate mortality across a population that may have moral status; this paper's narrower point is that the mortality p_make describes is homogeneous only in arithmetic, not in kind.

Second, Section 3's observation that each node reaches only a handful of viable neighbours, taken with this section's own admission that the nearest-targets rule ignores claiming and competition between branches, exposes a commons problem internal to the lineage. Ordinary, non-rogue sibling branches — not misbehaving outliers, just the model's own successful offspring — will tend to converge on the same small set of high-viability stars, since a target's viability is a property of the star, not of which branch reaches it first. This is distinct from the galaxy-as-commons tension the ethics paper develops between the lineage and other civilizations: here the competition is the lineage against itself, and R_eff is computed throughout this paper as though each node's nearest targets were its own to claim uncontested. Whether the knife-edge survives once claiming and competition are modelled explicitly — a question for the fleet simulator, not this demographic one — is open; the distributive question of who gets the handful of good stars is open regardless of the answer.

9. Conclusion

This paper computes the four engineering budgets the analytical companion set up, evaluating the reproduction demographics across the real solar neighbourhood. Mass is dominated by the in-situ factory and spans bootstrap-package ($\sim 10^3$ kg) to full-factory ($\sim 10^7$ kg) scales; braking against the interstellar medium is feasible for plausible magnetic-sail sizes and, through $d = m/(2\rho A)$, is independent of cruise speed; resource closure binds on manufacturing capability rather than on matter or energy, whose margins are astronomical; and reproduction is a knife-edge, with a mean effective reproduction number that crosses unity between two and three offspring per node and that a ten-point swing in per-stage reliability flips between extinction and expansion. The engineering lesson is that the design's first obligation is to drive R_eff safely above one — the demographic budget is the binding one, and the others are comfortable by comparison. The models and an interactive dashboard accompany this paper so the result can be reproduced and explored; the natural next step is the higher-fidelity model that derives the per-leg and viability terms from first principles and evaluates them across the Galaxy, turning this first-order computation into a mapped boundary of the regime in which a self-replicating probe both closes and expands.

References

Andrews, D. G., & Zubrin, R. M. (1990). Magnetic sails and interstellar travel. *Journal of the British Interplanetary Society*, 43, 265–272.

Freeland, R. M. (2015). Mathematics of magsails. *Journal of the British Interplanetary Society*, 68, 306–323.

Freitas, R. A. (1980). A self-reproducing interstellar probe. *Journal of the British Interplanetary Society*, 33, 251–264.

Freitas, R. A., & Merkle, R. C. (2004). *Kinematic Self-Replicating Machines.* Landes Bioscience.

Frisch, P. C., Redfield, S., & Slavin, J. D. (2011). The interstellar medium surrounding the Sun. *Annual Review of Astronomy and Astrophysics*, 49, 237–279.

Gaia Collaboration; Vallenari, A., et al. (2023). Gaia Data Release 3: Summary of the content and survey properties. *Astronomy & Astrophysics*, 674, A1.

Gros, C. (2017). Universal scaling relation for magnetic sails: Momentum braking in the limit of dilute interstellar media. *Journal of Physics Communications*, 1, 045007.

Harris, T. E. (1963). *The Theory of Branching Processes.* Springer-Verlag (Grundlehren der mathematischen Wissenschaften, Band 119).

Hart, M. H. (1975). An explanation for the absence of extraterrestrials on Earth. *Quarterly Journal of the Royal Astronomical Society*, 16, 128–135.

Metzger, P. T., Muscatello, A., Mueller, R. P., & Mantovani, J. (2013). Affordable, rapid bootstrapping of the space industry and solar system civilization. *Journal of Aerospace Engineering*, 26(1), 18–29.

NASA (1982). *Advanced Automation for Space Missions* (R. A. Freitas & W. P. Gilbreath, Eds.; NASA Conference Publication 2255). Proceedings of the 1980 NASA/ASEE Summer Study.

Poston, D. I., Gibson, M. A., Sanchez, R., & McClure, P. R. (2020). Results of the KRUSTY nuclear system test. *Nuclear Technology*, 206(sup1), S89–S117.

Redfield, S., & Linsky, J. L. (2008). The structure of the local interstellar medium. IV. Dynamics, morphology, physical properties, and implications of cloud-cloud interactions. *The Astrophysical Journal*, 673, 283–314.

Tipler, F. J. (1980). Extraterrestrial intelligent beings do not exist. *Quarterly Journal of the Royal Astronomical Society*, 21, 267–281.

von Neumann, J., & Burks, A. W. (1966). *Theory of Self-Reproducing Automata.* University of Illinois Press.

Companion code: `engineering.py` implements all four budgets and computes the reproduction demographics across the catalogue; `engineering_dashboard.html` recomputes the reproduction result live and exposes every parameter as a control. This paper is the computational sequel to the first-order analytical paper "Engineering Closure"; together they form the quantitative layer beneath the series' conceptual vehicle, payload, and bootstrapping papers. The higher-fidelity model — per-leg reliabilities derived from subsystem physics, viability from catalogue astrophysics, and evaluation across the modelled Galaxy — is the identified next step.